

District 4 SPSS Comparison 2023 - 2025 Sales

Elliot Dean - Program Analyst IV
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

11 May 2026

Contents

Model I	1
Model II	2
Model III	3
Model IV	4
Model V	6
Model VI	7

Model I

Table 1: District 4 | LN_SPPSF Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1297	1286	11	0.4876	0.0990	0.0910	12.8330
R	1297	1286	11	0.4876	0.0989	0.0912	12.8330
% diff (R vs SPSS)	0.00%	0.00%	0.00%	-0.01%	-0.10%	0.22%	0.00%

Table 2: District 4 | Model I Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	4.004	4.004	0.034	0.034	NA	NA	119.189	119.189	0.000	0.001
smallest	0.249	0.249	0.062	0.062	0.126	0.126	4.005	4.005	0.000	0.001
fireplace	0.124	0.124	0.032	0.032	0.121	0.121	3.893	3.893	0.000	0.001
yb_1946_to_1960	0.135	0.135	0.032	0.032	0.119	0.119	4.274	4.274	0.000	0.001
central_air	0.128	0.128	0.034	0.034	0.103	0.103	3.793	3.793	0.000	0.001
small	0.124	0.124	0.044	0.044	0.094	0.094	2.800	2.800	0.005	0.005
hot_water	0.138	0.138	0.048	0.048	0.081	0.081	2.880	2.880	0.004	0.004
months_15to24	0.011	0.011	0.004	0.004	0.068	0.068	2.529	2.529	0.012	0.012
qabove_avg	0.292	0.292	0.132	0.132	0.059	0.059	2.216	2.216	0.027	0.027
half_duplex1story	0.163	0.163	0.095	0.095	0.046	0.046	1.706	1.706	0.088	0.088
qbelow_avg	-0.072	-0.072	0.038	0.038	-0.065	-0.065	-1.871	-1.871	0.062	0.062
yb_post_1990	-0.388	-0.388	0.071	0.071	-0.159	-0.159	-5.489	-5.489	0.000	0.001

Model II

Table 3: District 4 | LN_PRICE (No ECFs) Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1707	1688	19	0.5020	0.2710	0.2620	32.9650
R	1707	1688	19	0.5020	0.2706	0.2624	32.9646
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.01%	-0.15%	0.15%	-0.00%

Table 4: District 4 | Model II Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	11.009	11.009	0.045	0.045	NA	NA	242.488	242.488	0.000	0.001
ln_base_bldgsize_ratio	0.778	0.778	0.094	0.094	0.327	0.327	8.246	8.246	0.000	0.001
ln_landratio	0.443	0.443	0.075	0.075	0.133	0.133	5.921	5.921	0.000	0.001
fireplace	0.134	0.134	0.029	0.029	0.115	0.115	4.660	4.660	0.000	0.001
central_air	0.136	0.136	0.032	0.032	0.092	0.092	4.284	4.284	0.000	0.001
good	0.384	0.384	0.108	0.108	0.076	0.076	3.539	3.539	0.000	0.001
smallest	0.171	0.171	0.068	0.068	0.075	0.075	2.525	2.525	0.012	0.012
yb_1946_to_1960	0.098	0.098	0.030	0.030	0.074	0.074	3.251	3.251	0.001	0.001
months_1to14	0.009	0.009	0.003	0.003	0.066	0.066	2.997	2.997	0.003	0.003
hot_water	0.116	0.116	0.045	0.045	0.061	0.061	2.560	2.560	0.011	0.011
single_family2story	0.073	0.073	0.035	0.035	0.054	0.054	2.065	2.065	0.039	0.039
small	0.079	0.079	0.043	0.043	0.052	0.052	1.821	1.821	0.069	0.069
qabove_avg	0.299	0.299	0.133	0.133	0.048	0.048	2.239	2.239	0.025	0.025
ln_garagespaces	-0.077	-0.077	0.046	0.046	-0.036	-0.036	-1.661	-1.661	0.097	0.097
half_duplex1story	-0.187	-0.187	0.093	0.093	-0.045	-0.045	-2.018	-2.018	0.044	0.044
qbelow_avg	-0.066	-0.066	0.035	0.035	-0.052	-0.052	-1.868	-1.868	0.062	0.062
unsound	-0.373	-0.373	0.132	0.132	-0.060	-0.060	-2.832	-2.832	0.005	0.005
poor	-0.232	-0.232	0.079	0.079	-0.062	-0.062	-2.920	-2.920	0.004	0.004
fair	-0.180	-0.180	0.031	0.031	-0.121	-0.121	-5.727	-5.727	0.000	0.001
yb_post_1990	-0.372	-0.372	0.071	0.071	-0.123	-0.123	-5.249	-5.249	0.000	0.001

Model III

Table 5: District 4 | LN_PRICE Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1707	1676	31	0.4533	0.4090	0.3980	37.4710
R	1707	1676	31	0.4533	0.4094	0.3985	37.4794
% diff (R vs SPSS)	0.00%	0.00%	0.00%	-0.00%	0.10%	0.13%	0.02%

Table 6: District 4 | LN_PRICE Model Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	1.0090	1.1310	0.9170	1.2330	0.4120	
R	0.9842	1.1048	0.9079	1.2168	0.3981	-0.4030
% diff (R vs SPSS)	-2.46%	-2.32%	-0.99%	-1.31%	-3.38%	

Table 7: District 4 | Model III Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	11.031	11.031	0.045	0.045	NA	NA	245.389	245.381	0.000	0.001
ln_base_bldgsize_ratio	0.475	0.475	0.075	0.075	0.200	0.200	6.361	6.360	0.000	0.001
ecf4r410	0.307	0.307	0.038	0.038	0.177	0.177	7.992	7.988	0.000	0.001
ecf4r415	0.587	0.587	0.070	0.070	0.168	0.168	8.352	8.358	0.000	0.001
ecf4r414	0.497	0.497	0.063	0.063	0.161	0.162	7.937	7.941	0.000	0.001
ecf4r422	0.535	0.535	0.073	0.073	0.148	0.148	7.286	7.285	0.000	0.001
ecf4r417	0.307	0.306	0.046	0.046	0.145	0.145	6.681	6.673	0.000	0.001
central_air	0.102	0.102	0.028	0.028	0.069	0.069	3.592	3.596	0.000	0.001
fireplace	0.076	0.076	0.026	0.026	0.065	0.065	2.872	2.892	0.004	0.004
two_baths	0.252	0.253	0.086	0.086	0.062	0.062	2.919	2.933	0.004	0.003
single_family2story	0.076	0.077	0.032	0.032	0.057	0.057	2.363	2.367	0.018	0.018
months_1to14	0.007	0.007	0.003	0.003	0.056	0.056	2.268	2.262	0.023	0.024
ecf4r413	0.115	0.115	0.047	0.047	0.052	0.052	2.427	2.424	0.015	0.015
ln_landratio	0.145	0.143	0.069	0.069	0.043	0.043	2.098	2.069	0.036	0.039
good	0.216	0.208	0.098	0.098	0.043	0.041	2.191	2.119	0.029	0.034
months_15to24	0.007	0.007	0.004	0.004	0.040	0.040	1.712	1.715	0.087	0.087
yb_1946_to_1960	0.049	0.049	0.028	0.028	0.037	0.037	1.756	1.767	0.079	0.077
unsound	-0.227	-0.227	0.120	0.120	-0.036	-0.036	-1.901	-1.897	0.057	0.058
crawl	-0.155	-0.155	0.084	0.084	-0.041	-0.041	-1.837	-1.840	0.066	0.066
ecf4r421	-0.181	-0.182	0.105	0.105	-0.041	-0.041	-1.736	-1.742	0.083	0.082
ecf4r416	-0.300	-0.301	0.135	0.135	-0.043	-0.043	-2.219	-2.221	0.027	0.026
half_duplex1story	-0.186	-0.186	0.082	0.082	-0.045	-0.045	-2.270	-2.264	0.023	0.024
ecf4r412	-0.200	-0.201	0.085	0.085	-0.046	-0.046	-2.344	-2.348	0.019	0.019
qbelow_avg	-0.059	-0.059	0.033	0.033	-0.046	-0.046	-1.815	-1.819	0.070	0.069
poor	-0.192	-0.192	0.072	0.072	-0.051	-0.051	-2.658	-2.657	0.008	0.008
ext_other	-0.087	-0.086	0.037	0.037	-0.054	-0.053	-2.334	-2.300	0.020	0.022
ecf4r408	-0.133	-0.133	0.046	0.046	-0.060	-0.060	-2.901	-2.903	0.004	0.004
ecf4r401	-0.168	-0.168	0.053	0.053	-0.063	-0.063	-3.140	-3.140	0.002	0.002
ecf4r411	-0.173	-0.173	0.047	0.047	-0.075	-0.075	-3.699	-3.702	0.000	0.001
ecf4r419	-0.319	-0.320	0.086	0.086	-0.078	-0.079	-3.699	-3.705	0.000	0.001
fair	-0.182	-0.182	0.029	0.029	-0.123	-0.123	-6.376	-6.374	0.000	0.001
ecf4r402	-0.678	-0.681	0.088	0.088	-0.150	-0.151	-7.670	-7.704	0.000	0.001

Model IV

Table 8: District 4 | LN_PRICE (Inliers) Model Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1576	1547	29	0.3945	0.4810	0.4720	49.5060
R	1576	1547	29	0.3945	0.4814	0.4717	49.5152
% diff (R vs SPSS)	0.00%	0.00%	0.00%	-0.00%	0.08%	-0.06%	0.02%

Table 9: District 4 | LN_PRICE (Inliers) Model Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	0.9990	1.0920	0.9370	1.1660	0.3490	-0.3160
R	0.9824	1.0787	0.9300	1.1600	0.3465	-0.3122
% diff (R vs SPSS)	-1.66%	-1.21%	-0.75%	-0.52%	-0.71%	1.21%

Table 10: District 4 | Model IV Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	11.135	11.135	0.030	0.030	NA	NA	373.745	373.777	0.000	0.001
ln_base_bldgsize_ratio	0.486	0.485	0.068	0.068	0.213	0.213	7.095	7.089	0.000	0.001
ecf4r410	0.303	0.303	0.034	0.034	0.190	0.190	8.855	8.853	0.000	0.001
ecf4r414	0.543	0.544	0.057	0.057	0.190	0.190	9.601	9.612	0.000	0.001
ecf4r415	0.562	0.562	0.063	0.063	0.174	0.174	8.892	8.898	0.000	0.001
ecf4r422	0.576	0.576	0.069	0.069	0.163	0.163	8.291	8.292	0.000	0.001
ecf4r417	0.309	0.308	0.041	0.041	0.159	0.159	7.451	7.443	0.000	0.001
central_air	0.122	0.122	0.026	0.026	0.089	0.090	4.753	4.764	0.000	0.001
months_15to24	0.011	0.011	0.003	0.003	0.068	0.068	3.656	3.657	0.000	0.001
two_baths	0.296	0.300	0.098	0.098	0.061	0.062	3.032	3.071	0.002	0.002
single_family2story	0.071	0.071	0.028	0.028	0.056	0.057	2.507	2.518	0.012	0.012
fireplace	0.061	0.061	0.024	0.024	0.056	0.056	2.531	2.550	0.011	0.011
good	0.239	0.222	0.101	0.101	0.045	0.042	2.358	2.203	0.018	0.028
ecf4r413	0.091	0.090	0.043	0.043	0.044	0.044	2.108	2.104	0.035	0.036
ecf4r404	0.113	0.113	0.060	0.060	0.042	0.041	1.900	1.893	0.058	0.059
ln_landratio	0.129	0.126	0.063	0.063	0.040	0.039	2.047	1.998	0.041	0.046
ecf4r412	-0.201	-0.202	0.088	0.088	-0.043	-0.044	-2.283	-2.289	0.023	0.022
half_duplex2story	-0.235	-0.234	0.101	0.101	-0.045	-0.045	-2.314	-2.310	0.021	0.021
unsound	-0.378	-0.377	0.142	0.142	-0.050	-0.049	-2.665	-2.659	0.008	0.008
ecf4r408	-0.122	-0.122	0.041	0.041	-0.060	-0.060	-3.010	-3.017	0.003	0.003
poor	-0.265	-0.265	0.081	0.081	-0.061	-0.061	-3.280	-3.275	0.001	0.001
half_duplex1story	-0.262	-0.262	0.083	0.083	-0.062	-0.061	-3.143	-3.138	0.002	0.002
ecf4r421	-0.273	-0.273	0.084	0.085	-0.068	-0.068	-3.226	-3.227	0.001	0.001
qbelow_avg	-0.080	-0.080	0.030	0.030	-0.068	-0.068	-2.688	-2.691	0.007	0.007
ecf4r411	-0.179	-0.179	0.045	0.045	-0.078	-0.079	-3.982	-3.987	0.000	0.001
ecf4r419	-0.362	-0.362	0.083	0.083	-0.091	-0.091	-4.355	-4.363	0.000	0.001
ext_other	-0.141	-0.140	0.037	0.037	-0.092	-0.091	-3.841	-3.822	0.000	0.001
ecf4r401	-0.230	-0.230	0.049	0.049	-0.093	-0.093	-4.728	-4.731	0.000	0.001
fair	-0.165	-0.165	0.026	0.026	-0.119	-0.119	-6.309	-6.308	0.000	0.001
ecf4r402	-0.794	-0.800	0.090	0.090	-0.168	-0.169	-8.847	-8.922	0.000	0.001

Model V

Table 11: District 4 | LN_TASP Model I Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	1576	1548	28	0.3944	0.4810	0.4720	51.2550
R	1576	1548	28	0.3944	0.4810	0.4717	51.2463
% diff (R vs SPSS)	0.00%	0.00%	0.00%	0.01%	0.00%	-0.06%	-0.02%

Table 12: District 4 | LN_TASP Model I Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	0.9980	1.0920	0.9380	1.1650	0.3490	-0.3160
R	1.0695	1.1739	1.0112	1.1609	0.3468	-0.3118
% diff (R vs SPSS)	7.17%	7.50%	7.80%	-0.35%	-0.63%	1.33%

Table 13: District 4 | Model V Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	11.246	11.249	0.029	0.029	NA	NA	392.935	393.090	0.000	0.001
ln_base_bldgsize_ratio	0.486	0.485	0.068	0.068	0.213	0.213	7.099	7.092	0.000	0.001
ecf4r410	0.303	0.303	0.034	0.034	0.190	0.190	8.856	8.857	0.000	0.001
ecf4r414	0.543	0.544	0.057	0.057	0.190	0.190	9.607	9.616	0.000	0.001
ecf4r415	0.561	0.562	0.063	0.063	0.174	0.174	8.893	8.902	0.000	0.001
ecf4r422	0.576	0.576	0.069	0.069	0.163	0.163	8.292	8.296	0.000	0.001
ecf4r417	0.309	0.309	0.041	0.041	0.159	0.159	7.461	7.449	0.000	0.001
central_air	0.122	0.122	0.026	0.026	0.089	0.090	4.751	4.767	0.000	0.001
two_baths	0.296	0.300	0.098	0.098	0.061	0.062	3.031	3.072	0.002	0.002
single_family2story	0.071	0.071	0.028	0.028	0.056	0.057	2.505	2.519	0.012	0.012
fireplace	0.061	0.061	0.024	0.024	0.056	0.056	2.526	2.553	0.012	0.011
good	0.239	0.222	0.101	0.101	0.045	0.042	2.357	2.203	0.019	0.028
ecf4r413	0.091	0.090	0.043	0.043	0.044	0.044	2.110	2.105	0.035	0.035
ecf4r404	0.113	0.113	0.060	0.060	0.042	0.041	1.900	1.894	0.058	0.058
ln_landratio	0.129	0.126	0.063	0.063	0.040	0.039	2.048	1.999	0.041	0.046
ecf4r412	-0.201	-0.202	0.088	0.088	-0.044	-0.044	-2.286	-2.291	0.022	0.022
half_duplex2story	-0.235	-0.234	0.101	0.101	-0.045	-0.045	-2.318	-2.311	0.021	0.021
unsound	-0.376	-0.377	0.141	0.141	-0.049	-0.049	-2.661	-2.666	0.008	0.008
ecf4r408	-0.122	-0.122	0.040	0.040	-0.060	-0.060	-3.011	-3.018	0.003	0.003
poor	-0.265	-0.265	0.081	0.081	-0.061	-0.061	-3.282	-3.276	0.001	0.001
half_duplex1story	-0.262	-0.262	0.083	0.083	-0.061	-0.061	-3.141	-3.140	0.002	0.002
qbelow_avg	-0.080	-0.080	0.030	0.030	-0.068	-0.068	-2.694	-2.693	0.007	0.007
ecf4r421	-0.274	-0.273	0.084	0.084	-0.068	-0.068	-3.246	-3.237	0.001	0.001
ecf4r411	-0.179	-0.179	0.045	0.045	-0.078	-0.079	-3.980	-3.992	0.000	0.001
ecf4r419	-0.362	-0.362	0.083	0.083	-0.091	-0.091	-4.364	-4.367	0.000	0.001
ext_other	-0.141	-0.140	0.037	0.037	-0.092	-0.091	-3.842	-3.824	0.000	0.001
ecf4r401	-0.230	-0.230	0.049	0.049	-0.093	-0.093	-4.729	-4.732	0.000	0.001
fair	-0.165	-0.165	0.026	0.026	-0.118	-0.119	-6.310	-6.317	0.000	0.001
ecf4r402	-0.793	-0.800	0.090	0.090	-0.168	-0.169	-8.848	-8.926	0.000	0.001

Model VI

Table 14: District 4 | LN_TASP Model II Fit Statistics: SPSS vs R

metricsource	df_total	df_resid	df_reg	see	r2	adj_r2	f_stat
SPSS	2042	2012	30	0.3953	0.4820	0.4750	62.5080
R	1576	1549	27	0.3942	0.4812	0.4721	53.2048
% diff (R vs SPSS)	-22.82%	-23.01%	-10.00%	-0.28%	-0.17%	-0.61%	-14.88%

Table 15: District 4 | LN_TASP Model II Assessing Statistics: SPSS vs R

metricsource	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
SPSS	0.9930	1.0790	0.9290	1.1620	0.3450	-0.3050
R	1.0733	1.1738	1.0110	1.1611	0.3450	-0.3111
% diff (R vs SPSS)	8.09%	8.79%	8.82%	-0.08%	-0.01%	-2.01%

Table 16: District 4 | Model VI Coefficient Comparison: SPSS vs R

term	b		Std. Error		Beta		t		Sig	
	R	SPSS	R	SPSS	R	SPSS	R	SPSS	R	SPSS
(constant)	9.570	9.970	0.249	0.164	NA	NA	38.451	60.864	0.000	0.001
lnsbldsfint	0.261	0.192	0.039	0.026	0.249	0.195	6.679	7.379	0.000	0.001
ecf4r414	0.548	0.546	0.057	0.053	0.192	0.178	9.687	10.277	0.000	0.001
ecf4r410	0.303	0.258	0.034	0.030	0.191	0.159	8.919	8.511	0.000	0.001
ecf4r415	0.566	0.561	0.063	0.060	0.176	0.160	8.967	9.383	0.000	0.001
ecf4r422	0.576	0.556	0.069	0.057	0.163	0.168	8.309	9.693	0.000	0.001
ecf4r417	0.313	0.297	0.041	0.037	0.162	0.151	7.584	8.019	0.000	0.001
central_air	0.122	0.109	0.026	0.024	0.090	0.075	4.772	4.548	0.000	0.001
two_baths	0.335	0.431	0.097	0.090	0.069	0.082	3.473	4.780	0.001	0.001
single_family2story	0.074	0.077	0.028	0.025	0.059	0.060	2.611	3.109	0.009	0.002
fireplace	0.062	0.053	0.024	0.021	0.057	0.048	2.618	2.518	0.009	0.012
good	0.248	0.210	0.101	0.086	0.047	0.041	2.452	2.444	0.014	0.015
ecf4r413	0.095	0.106	0.043	0.037	0.046	0.052	2.207	2.837	0.027	0.005
smallest	0.093	NA	0.055	NA	0.044	NA	1.705	NA	0.088	NA
ln_landratio	0.134	0.132	0.063	0.054	0.042	0.042	2.122	2.421	0.034	0.016
ecf4r404	0.112	NA	0.059	NA	0.041	NA	1.906	NA	0.057	NA
ecf4r412	-0.201	-0.202	0.087	0.070	-0.043	-0.048	-2.294	-2.877	0.022	0.004
unsound	-0.384	-0.319	0.141	0.116	-0.050	-0.045	-2.721	-2.755	0.007	0.006
ecf4r408	-0.119	-0.104	0.040	0.036	-0.059	-0.051	-2.948	-2.875	0.003	0.004
qbelow_avg	-0.072	-0.080	0.030	0.026	-0.061	-0.068	-2.398	-3.062	0.017	0.002
poor	-0.271	-0.237	0.081	0.068	-0.063	-0.056	-3.366	-3.477	0.001	0.001
ecf4r421	-0.285	-0.297	0.084	0.075	-0.071	-0.070	-3.381	-3.940	0.001	0.001
ecf4r411	-0.172	-0.199	0.045	0.037	-0.076	-0.095	-3.848	-5.404	0.000	0.001
ext_other	-0.127	-0.095	0.036	0.028	-0.083	-0.064	-3.537	-3.385	0.000	0.001
ecf4r401	-0.225	-0.271	0.049	0.043	-0.091	-0.109	-4.629	-6.336	0.000	0.001
ecf4r419	-0.373	-0.409	0.083	0.067	-0.094	-0.110	-4.500	-6.116	0.000	0.001
fair	-0.165	-0.174	0.026	0.022	-0.118	-0.128	-6.323	-7.807	0.000	0.001
ecf4r402	-0.786	-0.782	0.090	0.080	-0.166	-0.161	-8.770	-9.748	0.000	0.001
ecf4r403	NA	-0.083	NA	0.047	NA	-0.030	NA	-1.752	NA	0.080
ecf4r416	NA	-0.285	NA	0.142	NA	-0.033	NA	-2.008	NA	0.045
garage	NA	0.044	NA	0.022	NA	0.034	NA	2.013	NA	0.044
powder_room	NA	0.084	NA	0.030	NA	0.057	NA	2.794	NA	0.005
yb_1946_to_1960	NA	0.052	NA	0.022	NA	0.041	NA	2.347	NA	0.019